MOCO: Measure Once, Compile Once

Fraud detection on a quantum budget

Measure once	2 circuit executions per training transaction, in total
Train convexly	global optimum in PyTorch, no circuit inside the training loop
Noise cannot change the ranking	Spearman ρ = 1 at every noise level, by construction

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The baseline breaks under noise: AUPRC 0.580 to 0.405

492

frauds in 284,807 transactions (0.17%)

- A model that flags nothing is 99.83% accurate: accuracy says nothing. We rank by AUPRC (precision-recall).
- Common split, stratified 60/20/20, seed 42: test 56,962 rows with 98 frauds; threshold chosen on validation at the same noise level.

Baseline audit (kit VQC: ZZ map + EfficientSU2)

93
gates

33
CNOT

45
depth

126
noise locations

2,859,360 circuit executions to train and validate

Test AUPRC 0.580 ± 0.053 at p = 0 → 0.405 ± 0.177 at p = 0.05 (5 seeds)

Two facts from Task 1 make noise harmless for the ranking

Exercise 3.2: squashing keeps AUPRC

AUPRC depends only on the order of the scores. Compressing them towards 1/2 keeps the order, so AUPRC and AUROC do not change; only the threshold has to move.

$$s \;\to\; 1/2 + c\,(s - 1/2), \quad c > 0$$

Exercise 10.1: depolarizing scales ⟨Z⟩

A depolarizing channel after a one-qubit gate shrinks the Bloch vector by the same factor for every input.

$$\langle Z \rangle \;\to\; (1 - 4p/3)\,\langle Z \rangle$$

MOCO = a circuit where every qubit sees the same number of one-qubit gates and no CNOT, so the whole score is squashed by one factor λ^2 with $\lambda = 1 - 4p/3$.

Every choice is made on train or validation, never on test

- Encoder fitted on the training subset only: scale Time and Amount, PCA to 4 components, angles in [0, π].
- Undersample the training split, keeping all 295 training frauds; ratio and feature map chosen on validation (single 1:10: validation AUPRC 0.675 ± 0.012, 5 seeds).
- C of the convex fit chosen by 5-fold CV inside the training subset: no validation executions.
- Test set touched once, at the end; threshold maximises F1 on validation at the same noise level.

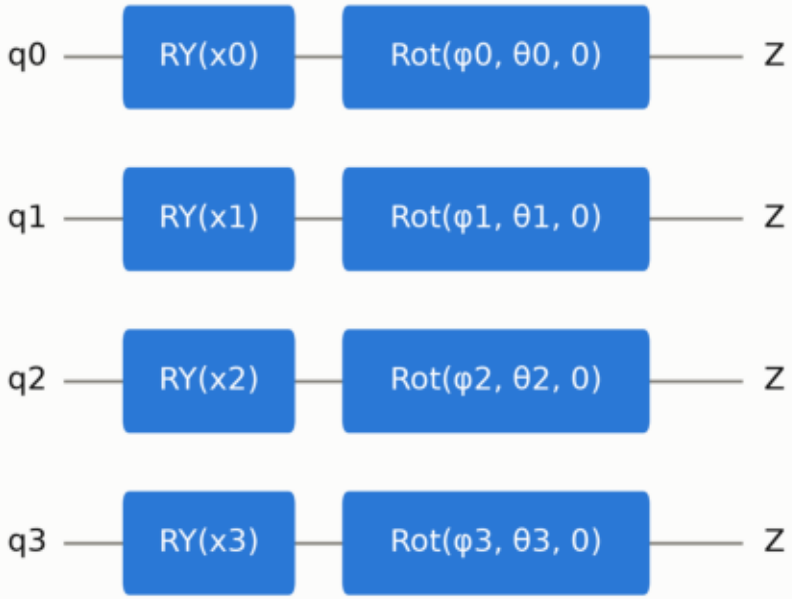
feature map	ratio	val AUPRC (5 seeds)	train exec.
single	1:1	0.564 ± 0.098	1,180
single	1:3	0.663 ± 0.020	2,360
single	1:10	0.675 ± 0.012	6,490
single	1:30	0.673 ± 0.018	18,290
single	1:100	0.648 ± 0.015	59,590
pair	1:1	0.623 ± 0.030	1,770
pair	1:3	0.635 ± 0.021	3,540
pair	1:10	0.662 ± 0.013	9,735
pair	1:30	0.675 ± 0.010	27,435
pair	1:100	0.631 ± 0.013	89,385

Chosen: single 1:10, best mean; pair 1:30 ties at 4x the executions.

MOCO has 8 noise locations, the baseline 126

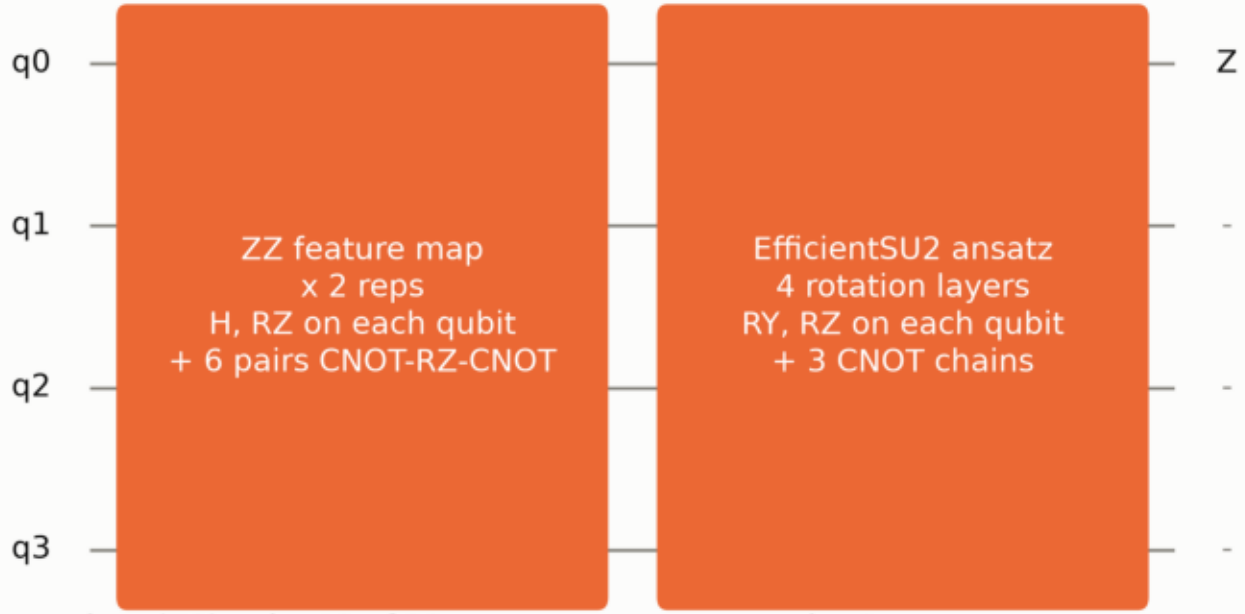
Every gate is followed by a depolarizing channel on each qubit it touches: 8 vs 126 noise locations

MOCO: 8 gates, 0 CNOT, 8 noise locations, depth 2



readout H = Σ w_q Z_q, p_fraud = (1 − ⟨H⟩) / 2

Baseline: 93 gates, 33 CNOT, 126 noise locations, depth 45



readout ⟨Z0⟩ only; 12 of 32 parameters never reach q0

Training the circuit is a convex fit on data measured once

1 Measure once

RY(x)|0⟩ has Bloch vector (sin x, 0, cos x). Read every training transaction in the X and Z bases: 2 executions per row, 6,490 in total.

2 Heisenberg picture

After a local Rot, $\langle Z_q \rangle = m_q \cdot r_q(x)$ exactly. The readout $H = \Sigma\, w_q Z_q$ is linear in the measured Bloch vectors.

3 Convex fit

Logistic regression on the measured vectors in PyTorch (LBFGS), C by CV on train: global optimum, no barren plateau, no circuit in the loop.

4 Compile back

Coefficients become Rot(−φ, −θ, 0) angles and observable weights; $p_{fraud} = (1 - \langle H \rangle)/2$. The circuit does the classification.

Same idea as classical surrogates of quantum models (Schreiber, Eisert, Meyer, PRL 2023) and data acquisition followed by classical processing (Cerezo et al., 2025): here the result is compiled back into the circuit.

Noise only squashes MOCO's scores: the ranking cannot change

1 Each depolarizing channel on a qubit shrinks its Bloch vector by $\lambda = 1 - 4p/3$ (exercise 10.1).

2 No CNOT and k = 2 gates on every qubit: the state stays a product, every $\langle Z_q \rangle$ is multiplied by λ^2.

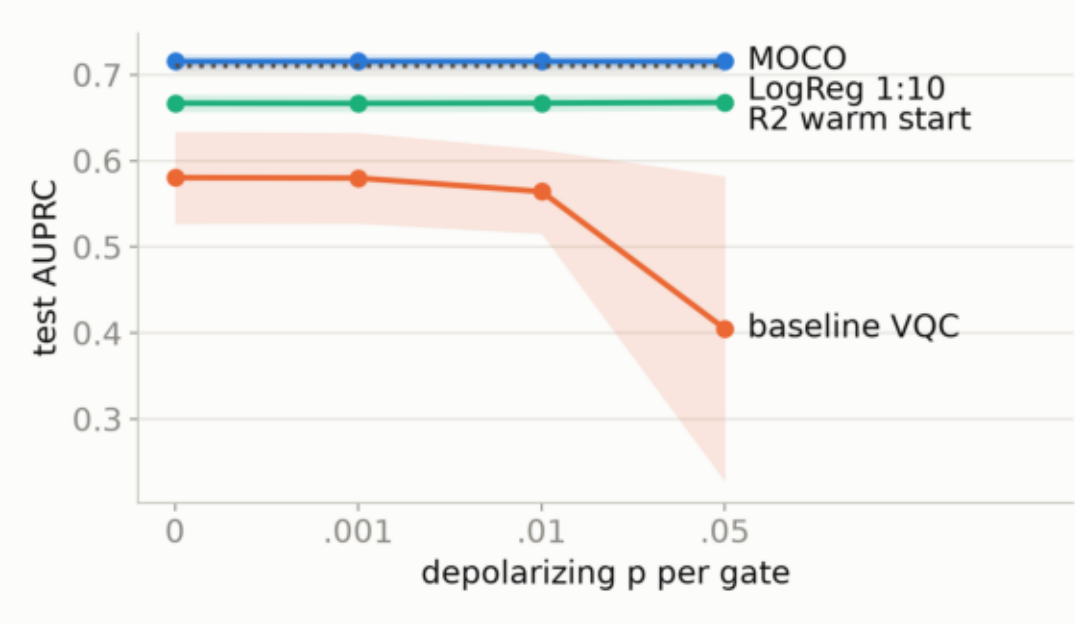
3 The readout is linear, so the whole score is squashed by λ^2 around 1/2 (exercise 3.2).

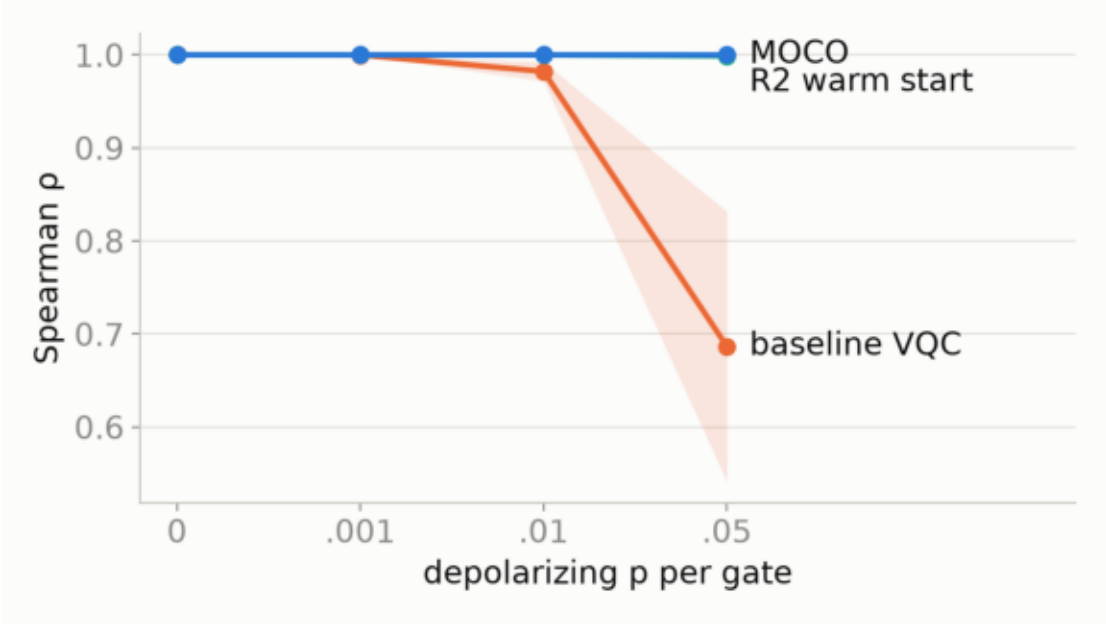
$$s_{noisy}(x) = 1/2 - \lambda^2 (1/2 - s_{ideal}(x))$$

- AUPRC, AUROC and $\rho = 1$ identical at every $p < 3/4$; the max-F1 threshold moves analytically: $t_p = 1/2 - \lambda^2 (1/2 - t_0)$.
- Checked against the kit's noise model (noisy_predict): max error 5.7e-14 on 200 rows at p = 0.05 and 0.5; effective compression equals λ^2 to 3e-17.

MOCO matches logistic regression and stays flat under noise

model (5 seeds)	AUPRC p=0	p=0.001	p=0.01	p=0.05	ρ .05	F1 .05	MCC .05
baseline VQC	0.580 ± 0.053	0.580 ± 0.053	0.564 ± 0.049	0.405 ± 0.177	0.686	0.537	0.545
R2 random init	0.635 ± 0.018	0.635 ± 0.018	0.636 ± 0.017	0.638 ± 0.015	0.999	0.760	0.760
R2 warm start	0.667 ± 0.010	0.667 ± 0.010	0.667 ± 0.010	0.668 ± 0.009	0.999	0.798	0.798
MOCO	**0.716 ± 0.007**	**0.716 ± 0.007**	**0.716 ± 0.007**	**0.716 ± 0.007**	**1.000**	**0.800**	**0.801**
LogReg 1:10	0.710 ± 0.007	0.710 ± 0.007	0.710 ± 0.007	0.710 ± 0.007	-	0.794	0.795
LogReg 1:1	0.703 ± 0.010	0.703 ± 0.010	0.703 ± 0.010	0.703 ± 0.010	-	0.791	0.791



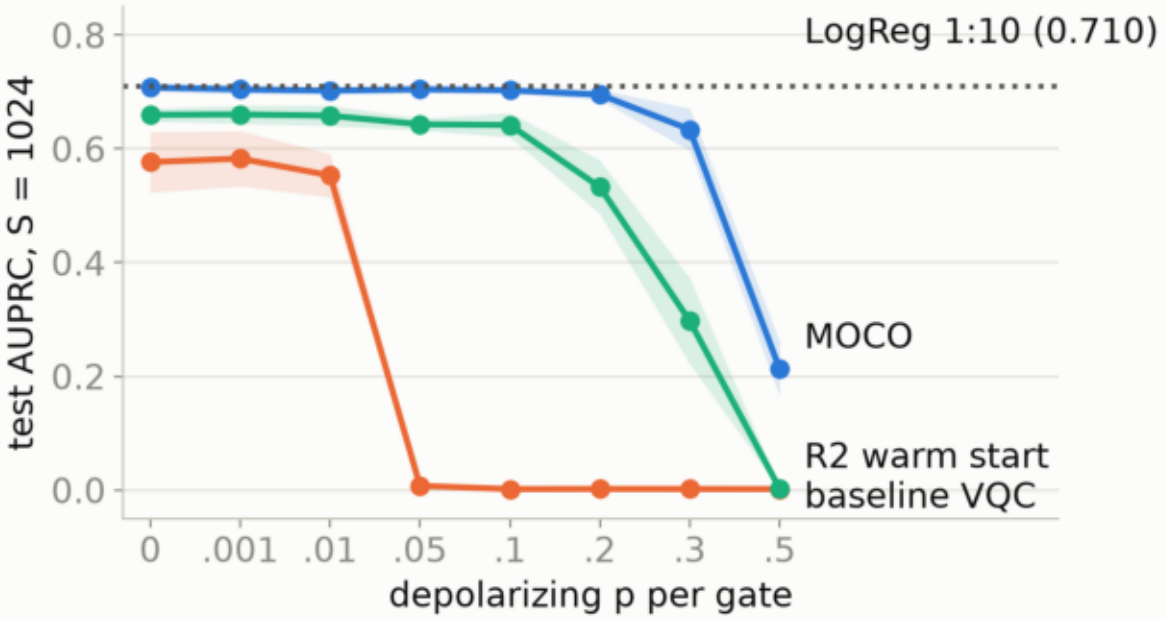


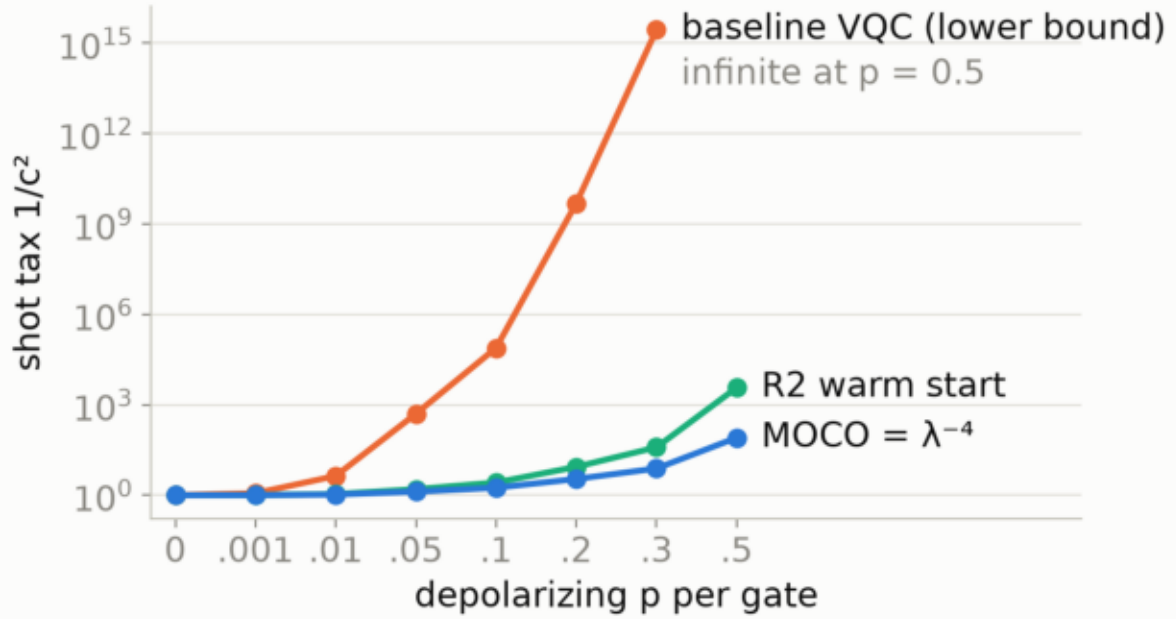
With real shots noise is a tax: MOCO 1.32x, baseline at least 504x

0.705
MOCO, test AUPRC with 1024 shots at p = 0.05

0.643
R2 warm start, same setting

0.007
baseline: chance level





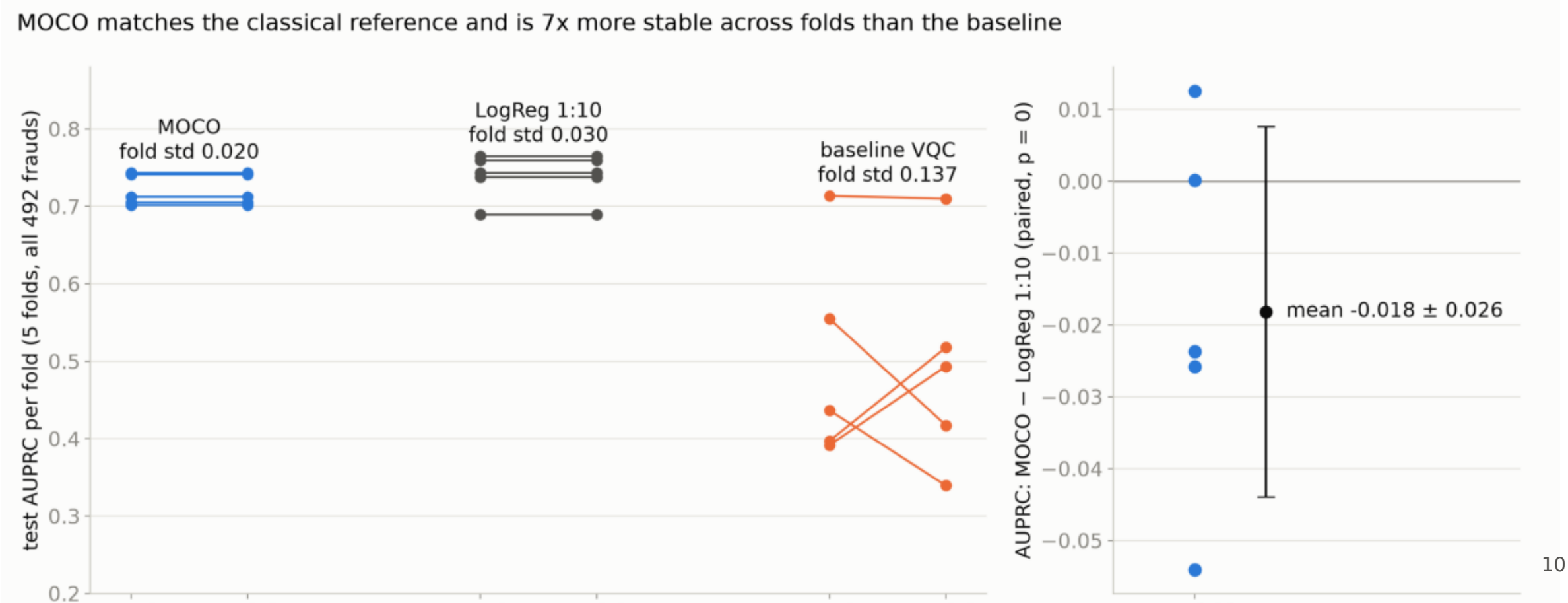
Shot tax $1/c^2$ at p = 0.05 (c = measured score compression): MOCO 1.32x (= λ^{-4}), R2 1.58x, baseline ≥ 504x (lower bound: its ranking is also lost).

All 492 frauds: MOCO matches LogReg and is 7x more stable

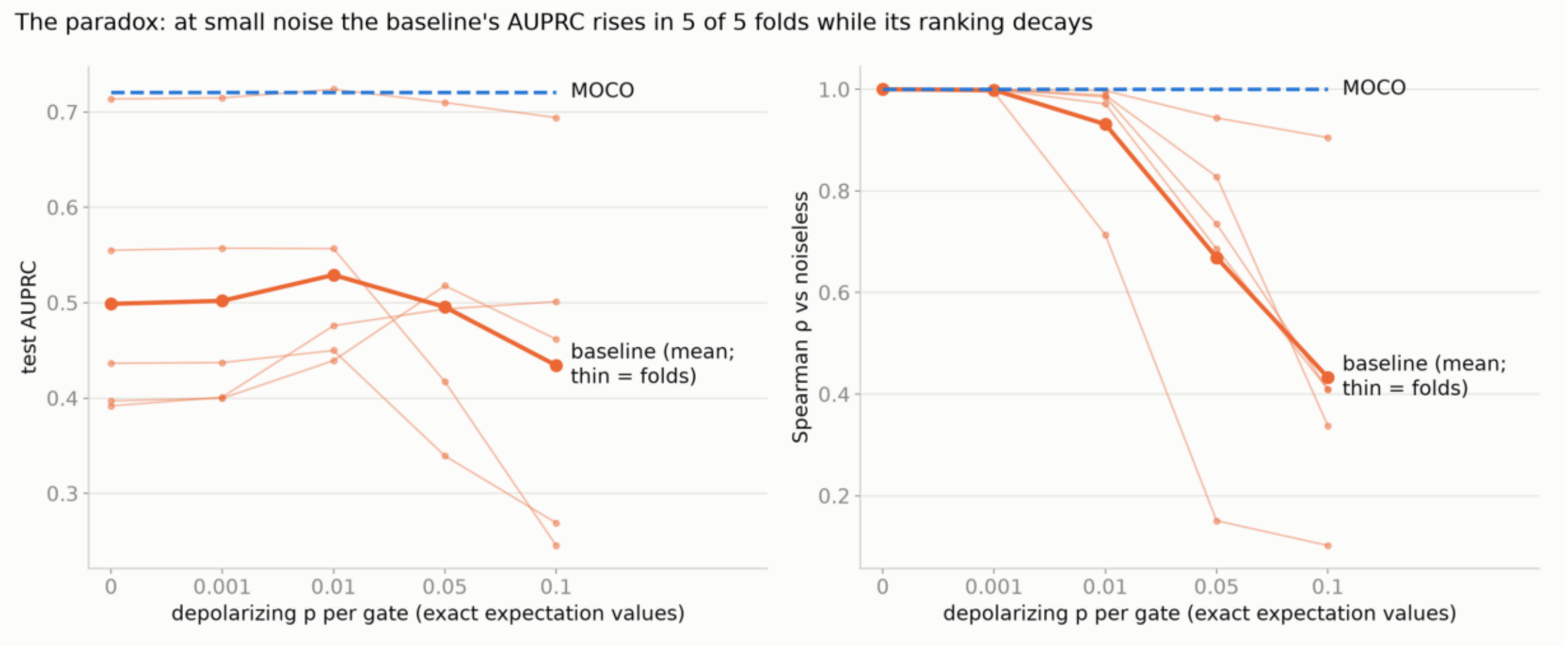
0.721 ± 0.020
MOCO, 5-fold CV, every p

0.739 ± 0.030
LogReg 1:10, same inputs

0.499 ± 0.137
baseline VQC, p = 0



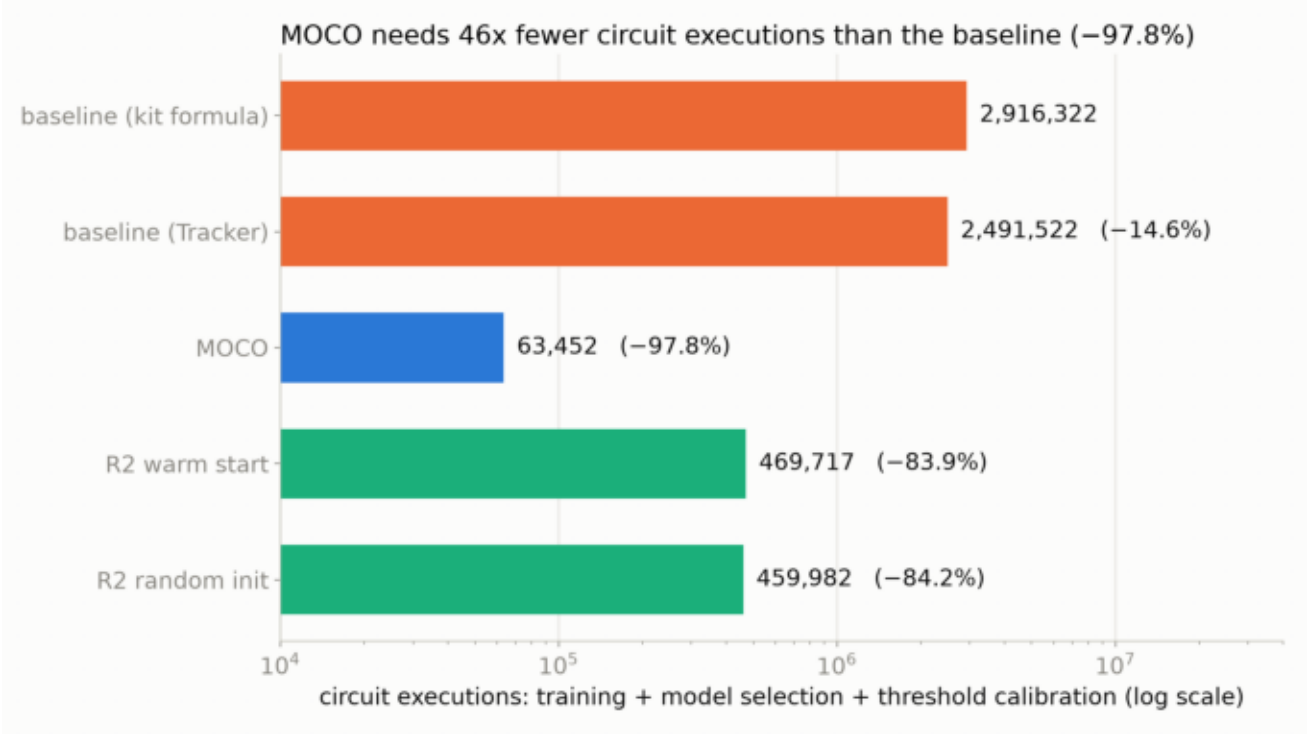
Paradox: the baseline's AUPRC rises while its ranking decays



AUPRC rises at p = 0.01 in 5 of 5 folds while ρ falls (ρ = 0.67 at p = 0.05): one metric can report an improvement where the model gets worse, as in the starting article. MOCO cannot show this: its ranking is invariant.

MOCO needs 46x fewer circuit executions

- Kit definition (training + model selection): baseline 2,859,360 vs MOCO 6,490 (-99.77%).
- With threshold calibration by the noise meter: baseline 2,916,322 vs MOCO 6,490 + [TBD: N_cal].
- Most conservative, threshold calibrated on the full noisy validation set: MOCO 63,452 (-97.8%).
- R2 warm start: 469,717 (-83.9%).



Adaptive triage needs a shot-aware encoding

Why triage fails on the kit encoding

Score margin between frauds and legit transactions [TBD: kit_margin] vs per-shot standard deviation [TBD: per_shot_std]: a few shots cannot separate them.

Robust percentile encoding

Angles from training-set percentiles keep the classes apart on the Bloch sphere. MOCO-robust test AUPRC at p = 0.05: [TBD: robust_auprc_p005].

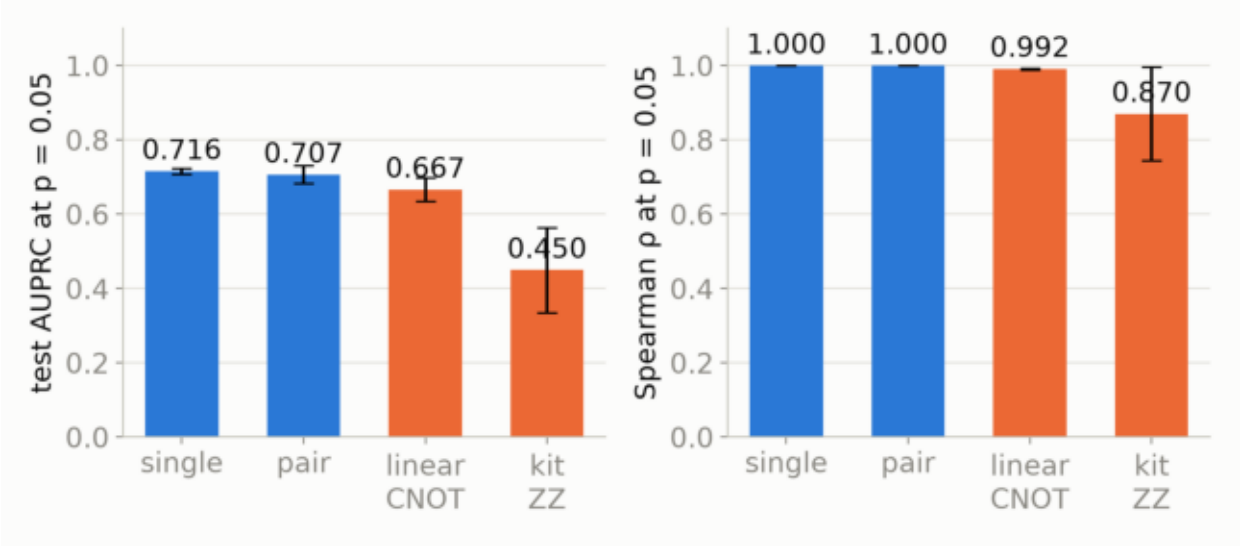
Noise meter and triage

Estimate p from [TBD: N_cal] calibration executions (error [TBD: noise_meter_p_error]), move the threshold analytically, stop sampling early: [TBD: triage_shots_per_tx] shots per transaction, F1 [TBD: triage_f1].

All numbers on this slide come from the adaptive-shots session (results/task2/adaptive_shots.csv, noise_meter.csv); red values are still being computed.

What did not work, measured honestly

- Light-cone pruning (R1): 15 gates removed, scores identical to 1e-10: zero gain.
- Entanglement: 3 CNOTs already lower AUPRC and break the invariance (figure).
- The kit's ZZ encoding caps at 0.44 even with the globally optimal readout.
- Noise-aware fine-tuning in R2: no gain over noiseless training.
- The shots-needed table was too noisy to use (validation and test agree in 24 of 40 cells).
- The baseline's drop under exact noise is fold-dependent (3 of 5 folds).



What we claim, what we do not, what comes next

We claim

- Robustness: $\rho = 1$ at every p, shot tax λ^{-4}.
- Stability: ±0.020 across folds (baseline ±0.137).
- Cost: 6,490 training executions (-99.77%).

We do not claim

- Quantum advantage: the circuit is a product state, classically simulable.
- Beating logistic regression: we match it (-0.018 ± 0.026 on all frauds).

Next: real hardware noise (amplitude damping, readout error); more qubits at zero robustness cost.

[1] Dinuț, Constantinescu, Alexandrescu, Electronics 15(11) 2489 (2026) · Schreiber, Eisert, Meyer, PRL 131, 100803 (2023) · Cerezo et al., Does provable absence of barren plateaus imply classical simulability?, arXiv:2312.09121 · Bowles, Ahmed, Schuld, arXiv:2403.07059 (2024) · Schuld et al., PRA 99, 032331 (2019) · Havlíček et al., Nature 567, 209 (2019) · Dal Pozzolo et al., IEEE SSCI (2015)